Uses, and misuses, of Automations in AAPS

Contribution to the discussion among DIY loopers

The authors assume no liability B.Loeff (Oct.25 publ. in AAPS Users FB) V.1.1

Certain Automations are likely to negatively affect the functioning of your AAPS loop

In this (rather long) post I would like to share my opinions about automations in AAPS, especially those that set a temp target or profile percentage based on BG value in order to let the algorithm work harder or less hard.

I see a lot of people making these kind of automations, intended to influence the working of their loop, e.g. by delivering more and bigger SMBs. Especially people who have just achieved objective 10 often go wild on these types of automations. In general I advise **against them as I don't think they will work and may even negatively affect the functioning** of your loop.

The main reason why I think this doesn't work is that the activity of insulin is much slower and longer than most people think. Even the fastest insulin on the market (Lyumjev) reaches a peak in its activity only after 45 minutes. For other insulin types this can even be up to 75 minutes. Suppose you use an automation to set a higher profile percentage when your blood sugar rises above a certain value, with the aim of getting more and larger SMBs. However the effect of those extra SMBs could take up to an hour to manifest. But by then your blood sugar has probably already started dropping, because AAPS has long since intervened. And those SMBs from an hour ago will mainly help to lower your blood glucose even more, resulting in the risk of a hypo.

The AAPS algorithm is in principle superior to what you might try via Automation

An important difference between automations and the AAPS algorithm is that automations can only respond to current situations, for instance your blood sugar level at this very moment. The AAPS algorithm takes decisions about actions like temporary basal rates and SMBs primarily based on predictions. To be able to do this AAPS uses knowledge about the activity graphs of the different types of insulins and has a well-functioning dynamic model for the absorption of carbohydrates (and decrease of COB). To me these predictions are the core of the system and the main reason why it works so well.

In fact with these types of automations you are trying to build your own loop algorithm. Which doesn't work because insulin simply isn't acting instantly and automations can't make predictions. In fact you are likely to make your BG graph into a rollercoaster this way, running from highs to lows and back again. Instead just let your closed loop do what it is designed for.

Advice to tune the AAPS loop *without* making use of Automations

Note: There is a reason why we "unlock" Automations late in the AAPS Objectives, after supposedly everybody has tuned their loop well.

My general strategy is simple and contains two pieces of advice:

First if you notice AAPS is not able to keep your BG values within target range don't reach for automations, but go back to basics. Start working on the basic settings in your profile: basal levels, carb ratios, correction factor (ISF) and DIA – in this order. Don't assume that the settings you had in your pump before looping are right – it quite often turns out they were not and AAPS is very sensitive to correct settings. The advice of doctors and diabetes nurses is often based on simple rules of thumb (like "the rule of 1500" and a fixed ratio between basal and bolus insulin) and is not suitable for closed looping. There is often only one really good method for finding the right basic settings: testing. You may have to stop your loop for half a day or a night and skip meals and boluses, but that won't kill you. It often yields a lot.

My second piece of advice is: lean back, sit on your hands and let your loop do its job. That's what it's designed for. Yes, I also have highs now and then, even in a (hybrid) closed loop they cannot always be avoided. I will resist the urge to intervene, because I know and I can see that my AAPS is already working hard to solve it. It may take a while, but I know my loop will do the job and in the end will provide a smooth landing without lows, which I rarely have. (But if you are sure your high is caused by something like a malfunctioning infusion site you shouldn't wait but take action of course).

Examples for beneficial Automations

There may be one or two exceptions to the rule. There is an automation that I think is useful, but it doesn't work the way most people think. When experiencing a low BG many people set a higher temp target for 30 minutes or so, whether manual or by an automation. I think this is even a standard feature in AAPS. This is a useful action, but does not really help to fight the low itself.
If your BG threatens to fall below 70 mg/dl or 4 mmol/l AAPS will have already reduced or stopped your basal rate in advance. Remember this is the only way for your loop to prevent hypos. Insulin that has already been delivered (and that probably has led to the low) cannot be removed from your body. The only solution that remains then is to eat some fast carbs, which of course you should do. As a result your BG will quickly rise again. But AAPS will react to this rise by delivering extra insulin (usually as SMBs) . The fast carbs you took will be absorbed in a short time, but the extra insulin will be active much longer. As a result you are likely to have another low in 30 to 45 minutes. To avoid this effect it is useful to set a high temp target for some time, so AAPS will not be too aggressive when your BG is rising after the low. In other words: this 'temp target when low' is not intended to solve the low itself, but to prevent the next one!

We can generalize a bit from this given example and say: As long as you let the loop work its usual way (based on predictions etc.) it *can be* helpful to use Automations *to modulate aggressiveness* (e.g. by setting temp. targets, by temp. modifying ISF). Carefully analyze your data - AFTER YOU DID A DECENT FIRST TUNING WITHOUT RESORTING TO AUTOMATIONS - to eventually see sets of Conditions where it might make sense to temporarily change aggressiveness. More see:
https://androidaps.readthedocs.io/en/latest/DailyLifeWithAaps/Automations.html#how-automation-can-help

In case you can *not* identify a reliable *automatic* trigger Condition, you can make it a "User Action" Automation => A grey button on the bottom of your AAPS main screen (for instance "Snack after sports") lets you select a pre-defined temp. mode with different aggressiveness, at your easy disposal whenever *you know* the relevant Condition (as reflected in the name you give your Automation, and on that grey button) is present. For more see https://androidaps.readthedocs.io/en/latest/AdvancedOptions/FullClosedLoop.html#fcl-s-automation-templates + examples given in Github/Bernie4375/FCL eBook, Case Studies 5.2 and 6.2 (these only work with autoISF dev version of AAPS, though)

Important: Always limit the *duration* for any Automations as much as you can. 3 Reasons for this: 1) You basically want to run your tuned loop. 2) Episodes with softer or stronger aggressiveness should be anchored on Conditions you see in your data. *As long as* these Conditions persist, the Automation would kick in again (=*automatically* prolong). 3) Any *other* Automation, even if Conditions arose that make it very desirable, is blocked while your prior Automation has not stopped. And when it has stopped, AAPS goes through all your Automations (those ticked as active=waiting to eventually kick in) strictly in the *sequence* of your list. So, make sure you list all your Automations in a reasonable order, so important ones are not shut out by less important ones! https://androidaps.readthedocs.io/en/latest/DailyLifeWithAaps/Automations.html#the-order-of-the-automations-in-the-list-matters

Beyond changing aggressiveness, Actions triggered by Automations might also set temporary boundaries

(insert added by Bernie)

Another exception could be if you want to do Full Closed Loop (FCL). I'm not doing this myself, but the chapter in AAPS Read the Docs about FCL states it should be done with the help of automations.

https://androidaps.readthedocs.io/en/latest/AdvancedOptions/FullClosedLoop.html#meal-detection-your-automations-for-boosting

Alternatives to AAPS Automations

For advanced users, and for users of other oref loops (Trio, iAPS), there are other possibilities to automate tasks using IFTTT, a third party Android app called Automate, or so-called Middleware.